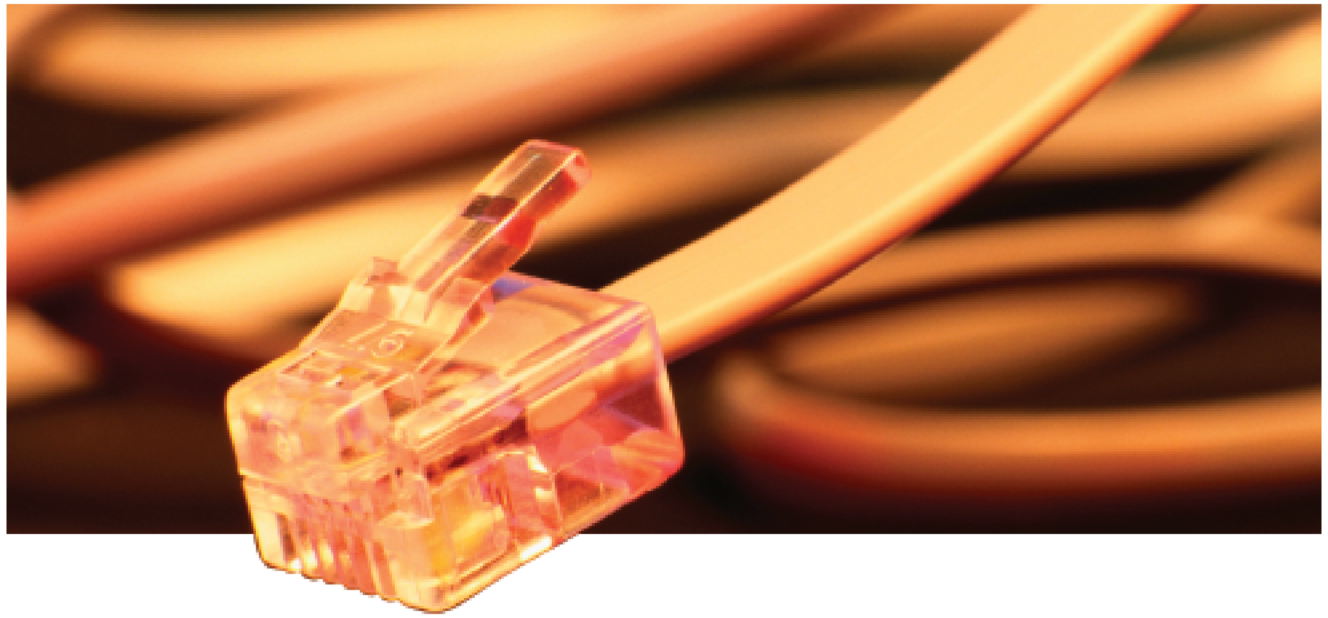

SIP On Call

The successful implementation of any IP telephony solution is a call control protocol that is widely accepted, easy to use and provides all the user features demanded of today's telecommunications services

The move to IP-based contact centers continues to gain momentum as organisations recognise the inherent benefits that can be gained by consolidating multiple contact channels into a single communication infrastructure. Moving from two separate networks to a single converged network provides immediate cost reductions in both staffing and maintenance of hardware and software. Also, deploying new services requires significantly lower investments in time and money than in traditional time-division multiplexing (TDM) systems.

The virtualisation capabilities of IP have resulted in significant cost savings and eliminated geographic limitations by allowing agents to sit at any PC desktop anywhere in the world. This location independence dramatically alters the contact center landscape. Above all else, however, an IP-based contact center facilitates the implementation of new cutting-edge applications that can deliver significant business gains in the areas of productivity, efficiency and customer satisfaction.

Building The Framework

Key to the successful implementation of any IP telephony solution is a call control protocol that is widely accepted, easy to use and provides all the user features demanded of today's telecommunications services. Emerging as a leading contender to fit this bill is the Session Initiation Protocol (SIP), an application-layer protocol for initiating, managing and terminating interactive sessions between one or more users on the Internet.

A key advantage of SIP is that it enables you to build new applications that use standard component-based equipment, which is less expensive and offers greater flexibility for scaling services compared to the TDM environment. Another SIP benefit—interoperability—is derived from the architecture of the Internet that has intelligence located at the edge of the network and devices communicating using standard protocols.

Organisations are beginning to recognise the inherent flexibility of SIP—particularly its ability to integrate well with the Web, email, streaming media applications and existing protocols used on the Internet.

Easily supporting a wide array of endpoint devices and configurations, SIP helps transform the communication paradigm from a device-centric to a user-centric model. Because of its flexible user location and name mapping features, SIP plays an important role in the delivery of rich mobility services such as personal mobility, call screening, forward and transfer, and multiparty calls.

Among the capabilities available in an SIP-enabled infrastructure include:

Unified communications—An SIP session can contain any combination of media (voice, data, video, etc.) and use different communication devices to help people communicate at any time and under their control.

IP-PBX functionality—An SIP-compliant IP-PBX can be used in a single office or multiple office environments, offering flexibility for future expansions.

Web IVR—SIP supports IVR features within a Web page environment, guiding users through options and providing auto-responses to common requests. Web IVR combines the rapid communications of voice with the ease of navigation of the Web to allow users to quickly scan the available links at any level.

Instant messaging (IM)—With support for instant messaging built into the SIP protocol, SIP makes it possible to promote an instant messaging session to a telephone call or video session at the click of a button. With integrated IM, agents can communicate with supervisors and experts while remaining on the call with the customer. This helps reduce talk times and allows agents to serve customers with greater speed, accuracy and efficiency.

Presence—The value of presence to a contact center is clear, and when coupled with voice, becomes even more compelling. Rather than presence being a simple "on" or "off," indicating the ability to receive instant messages, it can reflect a person's readiness and ability to communicate using a variety of different means.

Mobile phones and PDAs—Lightweight SIP client software can be embedded in mobile phones and PDAs so that new and existing services can cross all platforms. By extending intelligence out to devices like PDAs and cell phones, organisations will be able to create a virtual contact center that can effectively leverage a company's resources to their fullest.

The SIP standard only defines the signaling protocol for establishing and controlling sessions; it does not define how applications or features should be built, or how they should be delivered. This opens up the application development process, allowing applications from independent software vendors with expertise in specific vertical markets, spurring greater innovation.

SIP provides the foundation that allows you to do everything that you can do on TDM

Maximising The Capabilities Of IP

One of the most significant challenges facing organisations considering the move to IP is how to leverage existing systems. More specifically, they must be convinced that they will not only gain new features and functionality, but they will not lose the old applications contact center users enjoy with TDM and advanced CTI routing, including reporting and agent productivity applications. One way to overcome this challenge is to migrate to IP slowly.

The idea of location-independent services in the network means that a company can use a building-block approach for migrating its communications to IP on a site-by-site, group-by-group, or application-by-application basis. Because IP runs on a standards-based SIP infrastructure, it can be seamlessly integrated into the contact center, and organisations can easily make the transition in a phased approach, bringing value to the customer at each new phase.

Key to a successful phased roll-out is identifying those users who can show a near-term benefit in terms of improving business processes and targeting them for early implementation. This enables you to roll out new capabilities first to those people that will see the most immediate benefit, and migrate to others over time. For example, opening a new center in a low-cost region might present a more compelling proposition with IP technologies.

Each organisation needs to develop a migration strategy that maps closely to their overall goals and current infrastructure needs-one that can create a best-of-breed communication solution with reduced total cost of ownership and without sacrificing existing capabilities. The migration process can be initiated at any time and transitioned at whatever pace is desired. The good news is that the benefits of SIP can be leveraged even with a small initial deployment.

A key point to keep in mind when making a decision around IP technology is that it's not an all or nothing proposition-that is, there's no need to migrate your entire environment to a single-vendor IP solution to begin leveraging the advantages of IP. The beauty is that by choosing vendors that support an open standards-based approach, an enterprise can deploy multiple technologies from multiple vendors in different places and use SIP to enable all of these pieces to work together.

By designing a strategy around business needs and taking a phased approach, you can ensure that existing investments are not compromised, and that the implementation will not disrupt current business processes. SIP provides the foundation that allows you to do everything on SIP that you can do on TDM, while establishing the framework to build next-generation applications.

As an open protocol, SIP promotes interaction and involvement between individuals and companies in shaping how IP communications can grow and evolve. While many companies are looking at IP telephony as simply a cost-effective replacement to what they already have, its true value lies in the integration of IP technologies like Web, e-mail, and instant messaging and allowing these to become integral components of new voice services.

This openness of the protocol is expected to continue to drive development of new and innovative products and services. As long as applications that provide real business benefits running on SIP are available now, the technology has the potential to fulfill the vision that integrated communications has long promised. ⊠